

Question 3: Solar Irradiance Forecasting with Feature Engineering and CatBoost

Overview

You are given hourly weather and solar irradiance data for Bangkok from NASA POWER. Forecast the measured irradiance value for the next hour.

This problem requires building features that capture time periodicity, sun position, and weather conditions, along with summarizing the past 48 hours (mean, standard deviation, and lag) before feeding them into a CatBoost model.

Description

The data is split by time to simulate a genuine real-world forecast. The files provided are raw data from NASA POWER that has not been turned into features yet; you must build your own features (cyclic time, sun position, wind components, etc.) before feeding them into a model.

- `train.csv` : target time years 2019–2022, 35,063 rows, with raw columns and `target`
- `validation.csv` : target time year 2023, 8,760 rows, with raw columns and `target`
- `test.csv` : target time year 2024, 8,784 rows, with the same raw columns but no `target`

The raw columns from NASA POWER are `CLRSKY_SFC_SW_DWN` , `T2M` , `T2MDEW` , `RH2M` , `PS` , `WS10M` , `WD10M` , `PRECTOTCORR` , and `CLOUD_AMT` .

Every file has the same timestamp, target time, and feature columns. The test set has no target values, but you can concatenate the tail of validation with the start of test to build the 48-hour lookback without using future data.

Public Data

File structure under `data/public` :

```
train.csv
validation.csv
test.csv
```

`train.csv` and `validation.csv` have a `target` column, while `test.csv` does not. All three files are ordered by time and must not be shuffled before building lookback features.

Evaluation

Evaluated with MAE, RMSE, and R-squared, reporting both across all time periods and specifically for periods where the true irradiance value is greater than 20.

```
MAE = mean(abs(y_true - y_pred))  
RMSE = sqrt(mean((y_true - y_pred)^2))
```

Submission File

Submit `submission_solar.csv` with 8,784 rows.

```
id,predicted_ghi  
0,0.0  
1,0.0  
2,5.7
```

- `id` : the row order of `test.csv`
- `predicted_ghi` : the measured irradiance value for the next hour predicted by the model; must not be negative.



Leagues of Code